Differentiation: The Chain Rule

Questions 1–25

Question 1. Differentiation of $(ax^k + \ldots)^n$ Find the gradient function of:

a) $y = (2x^2 + 1)^3$
b) $y = (2x^3 - 5x)^4$
c) $y = (3x^4 - 1)^{1/2}$
d) $y = (1 - x^2)^{-3/2}$
e) $y = \frac{1}{5+\sqrt{x}}$
f) $y = (\frac{1}{x} - x)^7$

Question 2. Find the gradient of the curve $y = \sqrt{5x^2 - 2x + 1}$ at $(2, \sqrt{17})$

Question 3. Find the equation of the tangent to the curve $y = \sqrt{1 + 7x^2}$ when $x = 3$ giving your answer in the form $ay + bx = c$ where a, b and c are integers to be found.

Question 4. Find the equation of the normal to the curve $y = \frac{1}{\sqrt{4-3x^2}}$ when $x = -1$ giving your answer in the form $ay + bx = c$ where a, b and c are integers to be found.

Question 5. Find the coordinates of all the stationary points of each of the following curves:

a) $y = (3x^2 - 1)^4$

b) $y = \frac{1}{\sqrt{1+x^2}}$

Question 6. For each of the following, find $\frac{dy}{dx}$ giving your answer in terms of y:

a) $x = 3y^2 - y$

b) $x = 2y^4 - 1$

c) $x = (3y^2 - 1)^{1/4}$

Question 7. Find $\frac{dy}{dx}$ of each of the following at the given point:

a) $x = (3 - y)^5$ when $y = 2$

b) $x = (y^2 - 2y + 1)^{1/2}$ when $y = 4$

c) $x = \frac{1}{\sqrt{y^4+3}}$ for $y > 0$ when $x = \frac{1}{2}$

Question 8. Find the tangent of the curve $y = \frac{1}{3}(x^2 + 5)^{3/2}$ at $x = 2$ giving your answers in the form $y = mx + c$

Question 9. A curve has equation $x = (y-2)^6$

a) Find $\frac{dx}{dy}$ in terms of x

b) Hence find $\frac{dy}{dx}$

Question 10. Prerequisite knowledge for Q10: Points of Inflection Find the x-value of the point of inflection on the curve $f(x)$ given that $f'(x) = \frac{1}{(1+3x-2x^2)^3}$

Question 11. Differentiation of $\sin(\mathrm{f}(x))$ **&** $\cos(\mathrm{f}(x))$ Differentiate each of the following:

a) $y = \sin^2 x$
b) $y = \cos^2 x$
c) $y = 2\sin(3x - 7)$
d) $y = -\frac{1}{5}\cos(\pi x - \frac{\pi}{2})$
e) $y = \cos\sqrt{x}$
f) $y = -2\sin(x^3)$
g) $y = \cos(\sin x)$
h) $y = \sin(\sin x)$
i) $y = \sqrt{\sin x}$
j) $y = \sqrt[3]{\cos x}$
k) $y = 1 - \cos^3(5x)$

Question 12. Show that $\frac{d}{dx}[\frac{1}{\sin x}] = -\operatorname{cosec} x \cot x$

Question 13. A model for the tide height over a 24-hour period in Southampton is

$$x = 2.85 + 0.75 \sin\left(\frac{17\pi}{120}t + \frac{3\pi}{2}\right)$$

where the depth of the water is x metres and t is the number of hours passed since midnight.

a) State the maximum and minimum height the water reaches.

b) By using Calculus, find the two times at which it reaches its maximum depth in the first 24-hour period.

Question 14. A curve is given by $y = a\sin^2 x + b\cos^2 x$ where a and b are positive constants and are such that $a \neq b$. Show that the gradient function of the curve at $x = \frac{\pi}{4}$ is never zero.

Question 15. A model for the number of daylight hours, t (in hours), on the xth day of the year may be given by

$$t = 10 + 2.7\sin\left(\frac{2\pi}{365}(x - 78)\right)$$

Use this model to compare how the number of hours of daylight is changing on day 17 compared to on day 85.

Question 16. Show that the stationary points of $\mathrm{f}(x) = \sin(\frac{\pi}{2}e^x)$ occur when $x = \ln k$ where k is a positive odd integer.

Question 17. Show that $\mathrm{f}(x) = \frac{1}{2}x + \cos(\frac{2}{x})$ is an increasing function for all $x > 2$

Question 18. Differentiation of $e^{\mathrm{f}(x)}$, $\ln(\mathrm{f}(x))$, $a^{\mathrm{f}(x)}$ Differentiate each of the following:

a) $y = (e^{2x} - 1)^5$
b) $y = (5^x - 1)^4$
c) $y = (\ln x)^4$
d) $y = e^{3x^2-1}$
e) $y = 4^{x^2+5x+1}$
f) $y = -5^{2x^2+1}$

Question 19. Find coordinates of the stationary points on the following curves:

a) $y = 6x - e^{2x}$

b) $y = 2\ln(x - x^2)$

Question 20. Differentiate each of these:

a) $y = \ln(2x^3 + x^2 - 1)$

b) $y = \ln(\sqrt{x})$

c) $y = \ln(\frac{1}{\sqrt{x}})$

d) $y = \ln(\sin x)$

e) $y = \ln(e^x + 1)$

f) $y = \ln(\ln x)$

Question 21. Differentiation of $\tan(\mathrm{f}(x))$ Differentiate each of the following:

a) $y = \tan\sqrt{x}$

b) $y = \tan(\sin x - x^2)$

c) $y = e^{\tan x}$

d) $y = \tan(\ln x + e^{2x})$

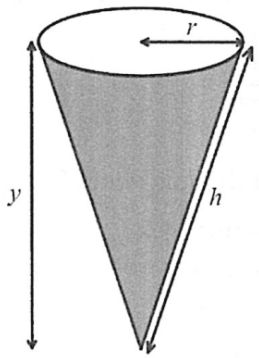
e) $y = \tan(\tan x)$

Question 22. Show that $\frac{d}{dx}[\ln(\tan x)] = \operatorname{cosec} x \sec x$

Question 23. A curve is given by $y = \tan(x^2)$ where $0 < x < \sqrt{\frac{\pi}{2}}$. Show that the curve is always increasing.

Question 24. A curve is given by $y = \tan^2(ax)$ where $0 < a \leq 2\pi$. Given that when $x = 1.5$, $\frac{dy}{dx} = 0$ find the possible values of a

Question 25. An ice cream cone is to be made from 30π cm^2 of wafer. The ice cream cone may be modelled with radius r cm, height y cm and slant height h cm and an assumption of negligible thickness.



Curved surface area: $A = \pi rh$
Volume: $V = \frac{1}{3}\pi r^2 y$

a) Show that the volume is given by $V = \frac{1}{3}\pi\sqrt{900r^2 - r^6}$
b) Hence, find the maximum volume of the cone.